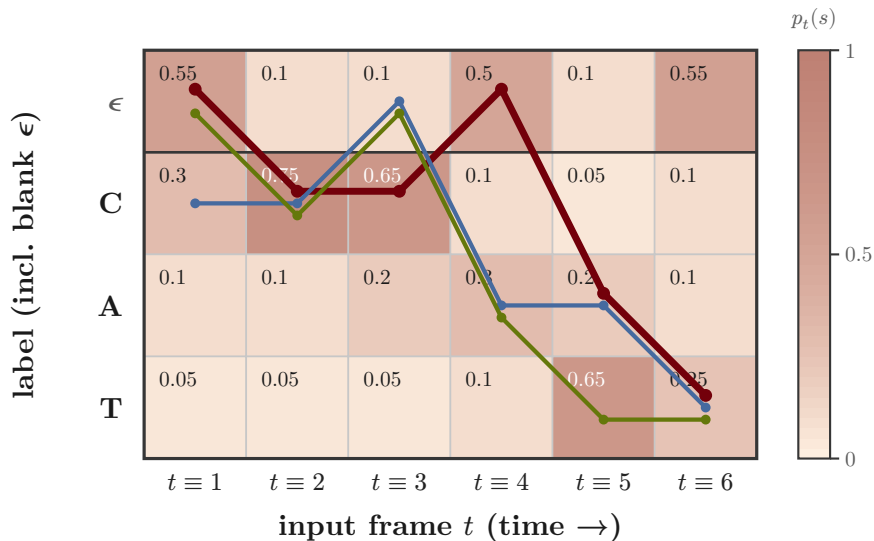
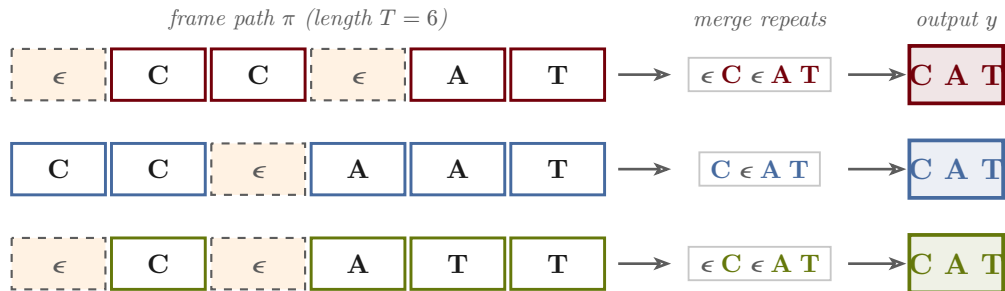


CTC alignment lattice and collapse

per-frame softmax over symbols $\cup \{\text{blank } \epsilon\}$; monotonic paths π pick one row per frame



collapse $B(\pi)$: merge adjacent repeats $\rightarrow$ drop blanks ϵ



three distinct alignments $\rightarrow$ one output $\rightarrow p(y) = \sum_{\pi \in B^{-1}(y)} \prod_t p_t(\pi_t)$